A car is speeding up over a time interval from $t = 0$ to $t = 4$ seconds. Values of the velocity are given in the table below for selected values of t:

Time (seconds)	0	1	2	3	4
Velocity (feet/sec)	12	13	16	21	28

Suppose we wish to estimate the distance the car traveled between $t = 0$ and $t = 4$ seconds. That is, we wish to find the displacement x given the velocity $v = \frac{dx}{dt}$.

1. Find an underestimate for the distance the car traveled between $t = 0$ and $t = 1$ seconds.

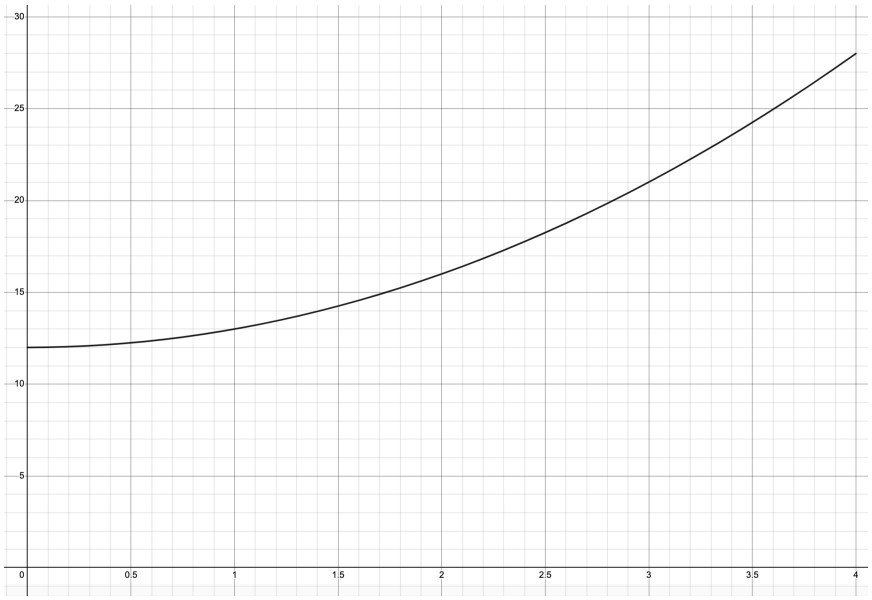
2. Find an overestimate for the distance the car traveled between $t = 0$ and $t = 1$ seconds.

3. Find an underestimate for the distance the car traveled between $t = 0$ and $t = 4$ seconds by repeating your method in Question 1 for each 1 second interval.

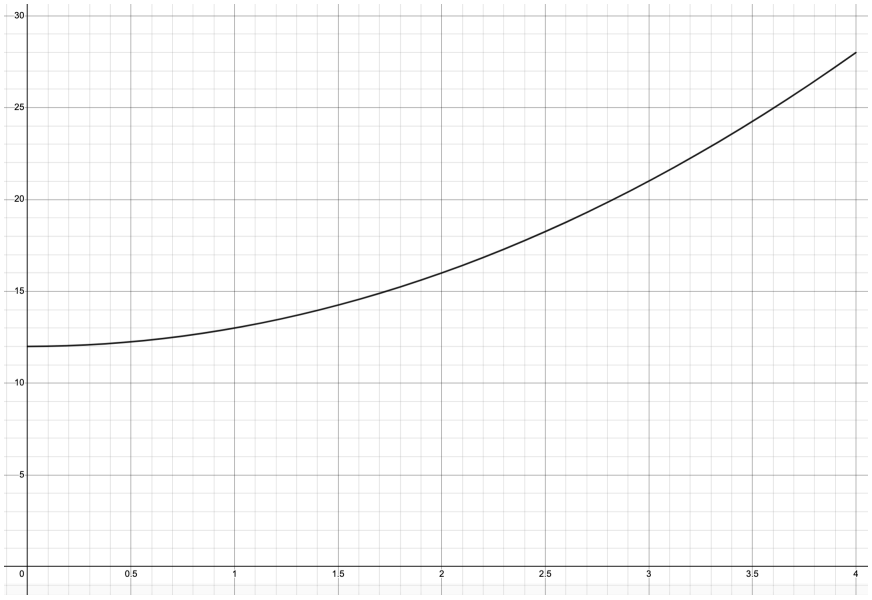
4. Find an overestimate for the distance the car traveled between $t = 0$ and $t = 4$ seconds by repeating your method in Question 2 for each 1 second interval.

We can interpret the equation $\Delta x = v\Delta t$ geometrically by viewing Δx as the area of a rectangle of height v and width Δt.

4. Suppose the exact velocity function is the function graphed below. Use the interpretation above to plot the steps of your underestimate from question 3 on the same axes.



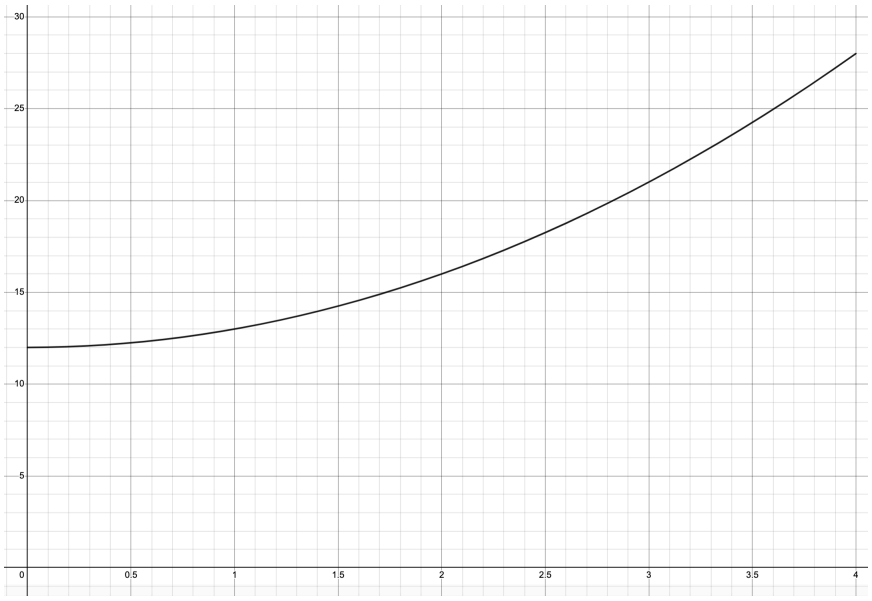
5. Now use the interpretation above to plot the steps of your overestimate from question 4 on the axes below.



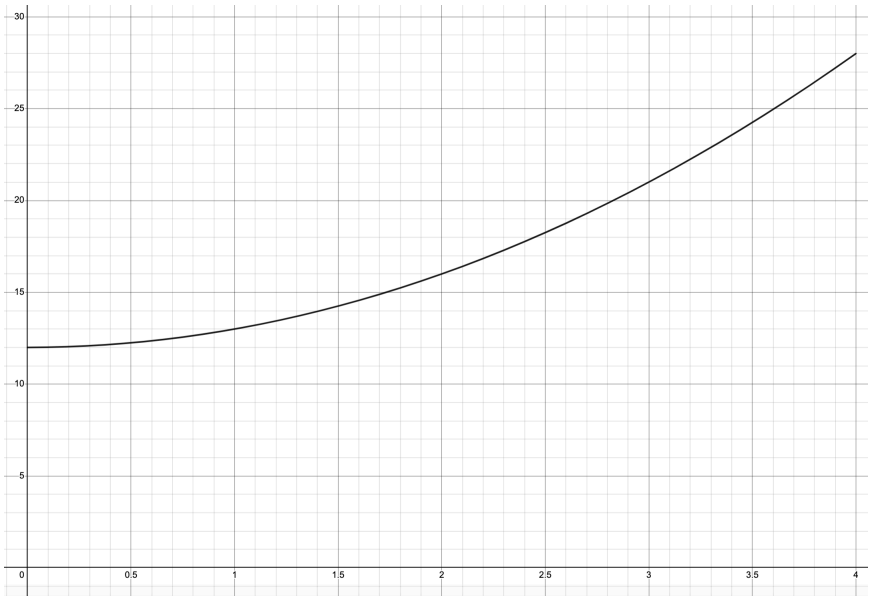
Now suppose more data is known:

Time (seconds)	0	0.5	1	1.5	2	2.5	3	3.5	4
Velocity (feet/sec)	12	12.25	13	14.25	16	18.25	21	24.25	28

6. Repeat your method used above to find an underestimate for the distance the car traveled between $t = 0$ and $t = 4$ seconds and plot the steps of your calculation on the axes below.



7. Repeat your method used above to find an overestimate for the distance the car traveled between $t = 0$ and $t = 4$ seconds and plot the steps of your calculation on the axes below.



8. How did your over- and underestimates change when you acquired more data?

9. In light of your answer to Question 8, what would you do to get a better estimate of the distance the car traveled between $t = 0$ and $t = 4$ seconds?

10. Using your graphs, how would you characterize geometrically the *exact* distance the car traveled between $t = 0$ and $t = 4$ seconds?